

Aging Mitigation Strategies for Heterogeneous Multi-Core Processors: A System Management Controller-Based Approach

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Abstract—The continuous advancement of transistor technology has enhanced integrated circuit performance while introducing reliability challenges due to aging mechanisms such as NBTI, HCI, TDDB, and Electromigration. This research investigates software, hardware, and hybrid techniques for mitigating aging in multi-core processors, including DVFS, power gating, clock gating, instruction scheduling, and task migration. We propose an adaptive, multi-level strategy integrated into a System Management Controller (SMC) that leverages real-time sensor data (temperature, voltage, aging) to dynamically adjust system parameters. Implemented on a RISC-V big.LITTLE architecture using RTL and TLM simulations, our framework ensures extended processor lifespan and reliability with minimal performance overhead. Future work targets enhancing decision algorithms and optimizing mitigation across diverse workloads.

Index Terms—Aging mitigation, multi-core processors, big.LITTLE architecture, Reliability, RISC-V, RTL simulation, TLM simulation, System Management Controller (SMC).

I. INTRODUCTION

The rapid advancement of semiconductor manufacturing technologies has enabled the integration of billions of transistors on a single chip, significantly enhancing the performance and energy efficiency of modern multi-core processors. This evolution, known as technology scaling (mise en échelle de la technologie), has led to smaller transistor geometries, higher switching speeds, and lower supply voltages.

However, as transistor dimensions shrink below the nanometer scale, reliability challenges become increasingly prominent. One of the most critical among these is circuit aging, a set of degradation mechanisms that gradually deteriorate transistor characteristics over time. These mechanisms, including Negative Bias Temperature Instability (NBTI), Hot Carrier Injection (HCI), Time-Dependent Dielectric Breakdown (TDDB), and Electromigration, can cause shifts in threshold voltage, increased path delays, and ultimately, system failures.

In contrast to traditional reliability issues such as manufacturing defects or transient faults, aging effects are progressive and predictable, making them suitable targets for proactive mitigation. Yet, if not properly addressed, aging-induced degradations can significantly impact the lifetime, performance, and energy efficiency of integrated systems.

Scientific Objectives: This research aims to explore and evaluate a range of aging mitigation techniques specifically adapted to multi-core processors. The primary objectives are:

1) To analyze the physical mechanisms of aging and their effects on performance and reliability, 2) To review and compare existing online countermeasure strategies at software, hardware, and hybrid levels and 3) To propose a real-time, sensor-driven framework capable of dynamically adapting to aging conditions with minimal architectural changes.

By addressing the aging challenge through intelligent monitoring and control, this work contributes to the design of more reliable, longer-lasting multi-core systems suitable for a wide range of applications—from high-performance servers to energy-constrained mobile devices.

Paper Organization: The remainder of this paper is organized as follows:

- Section 2 provides a detailed overview of the main aging mechanisms affecting transistor reliability.
- Section 3 presents a comprehensive analysis of existing online countermeasure techniques, including their operational principles and trade-offs.
- Section 4 discusses simulation-based evaluation platforms used to assess the effectiveness of aging mitigation strategies.
- Section 5 offers a comparative analysis of the techniques in terms of aging coverage, performance impact, and implementation complexity.
- Section 6 describes the proposed multi-level mitigation framework based on the System Management Controller (SMC), applied to a big.LITTLE heterogeneous architecture.
- Section 7 concludes the paper and outlines directions for future research.

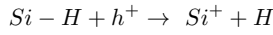
II. BACKGROUND AND AGING MECHANISMS

As semiconductor devices scale down to nanometer dimensions, reliability concerns become increasingly critical. Among these concerns, aging mechanisms such as NBTI, HCI, TDDB, and EM significantly impact the performance and lifespan of integrated circuits. This section provides an overview of these

mechanisms, their underlying physics, and their effects on device reliability.

A. Negative Bias Temperature Instability (NBTI):

NBTI primarily affects p-channel MOSFETs (pMOS) when a negative voltage is applied to the gate. Over time, this leads to the generation of interface traps at the silicon-dielectric interface and charge trapping in the gate dielectric, resulting in an increase in the threshold voltage (V_{th}) and a decrease in carrier mobility [1]. This degradation manifests as slower transistor switching speeds and reduced drive current. When a pMOS transistor is subjected to negative gate bias, holes accumulate at the Si/SiO₂ interface, leading to the breaking of Si-H bonds and the formation of interface traps. The reaction can be represented as:



where h^+ denotes a hole.

The accumulation of interface traps and trapped charges in the dielectric layer causes a shift in V_{th} , degrading the transistor's performance over time. This degradation is particularly significant in circuits that remain in the 'on' state for extended periods.

The degradation of V_{th} due to NBTI can be modeled as [2], [3]:

$$\Delta V_{th} = C_{(N/PBTI)} \times \exp(\beta |V_{gs}|) \times \exp\left(-\frac{E_{a(N/PBTI)}}{kT}\right) \times t^n$$

B. Hot Carrier Injection (HCI):

HCI is a critical aging mechanism that predominantly affects nMOS transistors during switching events, specifically when the transistor operates in saturation mode. In this state, high lateral electric fields near the drain region accelerate charge carriers (primarily electrons), causing them to gain significant kinetic energy.

When these hot carriers acquire enough energy, they can overcome the energy barrier and become injected into the gate dielectric. This injection leads to the formation of permanent positive charges in the oxide layer and interface traps at the silicon-oxide interface. As a result, the transistor experiences a shift in threshold voltage (V_{th}), degradation in channel mobility and an overall deterioration in current drive capability and switching performance [4].

Interestingly, the severity of HCI degradation decreases at higher temperatures due to several physical effects: 1) The carrier saturation velocity region shortens, reducing the energy gained by electrons, 2) The mean free path of majority carriers is reduced, lowering the likelihood of high-energy collisions, 3) The impact ionization process becomes less efficient, thus limiting hot carrier generation.

The net effect of these phenomena is a reduced probability of hot carriers penetrating the oxide at elevated temperatures, making HCI less dominant in thermally stressed environments compared to mechanisms like NBTI.

The lifetime (t_f) of a device under HCI stress is commonly modeled by [5]:

$$t_f = A_{HCI} \left(\frac{I_{sub}}{W}\right)^{-n} \exp\left(\frac{E_{aHCI}}{kT}\right)$$

This model shows how HCI degradation is highly dependent on electric field conditions and moderately on temperature, with increased current levels accelerating the degradation process.

The following table summarizes the key differences between these mechanisms regarding causes, effects, and acceleration factors.

TABLE I
SUMMARY OF DEGRADATION MECHANISMS AND THEIR CONDITIONS

Mechanism	Voltage	Temp.	Freq.	Duty	Observation
BTI	High	High	Low	High	Deterministic and progressive
HCI	High	Low	High	—	Deterministic and progressive
TDDDB	High	High	Low	High	Random and sudden
Electromigration	Medium	Very High	High	—	Random and abrupt

As aging effects continue to compromise the reliability, performance, and power efficiency of integrated circuits over time, the development and deployment of effective aging countermeasure techniques have become essential. These techniques are critical not only to increase the Mean Time To Failure (MTTF) and extend the systems operational lifetime, but also to preserve acceptable levels of performance and power consumption throughout the circuit's usage span.

Two main classes of countermeasure approaches have emerged:

- **Offline Techniques:** These are typically applied during the design phase. Offline techniques involve optimizing the architecture, layout, or timing of circuits based on predefined workload scenarios and aging stress models. While effective for static environments, offline techniques cannot account for runtime variability or unforeseen stress conditions that emerge during actual use.
- **Online Techniques:** In contrast, online countermeasure techniques operate dynamically, monitoring the circuit's real-time conditions (e.g., temperature, voltage, workload patterns) and adapting the system configuration accordingly. These techniques are increasingly vital for modern multi-core systems, where dynamic workloads, varying execution modes, and heterogeneous core usage introduce substantial variability in aging behavior.

Online countermeasures provide the flexibility to mitigate aging effects in real-time, ensuring system dependability across diverse usage conditions. In the following section, we examine various online techniques that have been developed to address aging in multi-core processors.

III. ONLINE AGING COUNTERMEASURE TECHNIQUES

A. Power Gating

Power Gating is a widely adopted technique used to reduce leakage power in digital circuits, especially when the system or submodules are idle. It works by introducing a high-threshold voltage sleep transistor between the logic block and the power supply rail (VDD or GND). When the circuit enters sleep mode, the sleep transistor disconnects the block from the power rail, minimizing leakage energy dissipation.

In aging mitigation contexts, PG is effective against Negative Bias Temperature Instability (NBTI) by reducing the stress time on transistors, thus lowering the aging rate. When implemented with awareness of NBTI dynamics, this technique significantly extends the system's lifetime while concurrently saving energy [6].

The proposed method involves inserting a sleep transistor (ST) either at the ground or power supply node of logic gates. When sleep is enabled, the supply is virtually disconnected, creating a low-leakage condition. The size of the ST directly influences area, power, and delay: A smaller ST reduces area and leakage but increases delay. A larger ST decreases delay but consumes more area and power.

The impact of NBTI on delay without PG is expressed as:

$$d'(t) = d \cdot \left(1 + \frac{(k \cdot (\beta \cdot t)^n)}{(V_{GT} - k \cdot (\beta \cdot t)^n)} \right)$$

With PG, it becomes:

$$d'_s(t) = d_s \cdot \left(1 + \frac{(k \cdot (\beta \cdot (1 - P_{sleep}) \cdot t)^n)}{V_{GT} - k \cdot (\beta \cdot (1 - P_{sleep}) \cdot t)^n} \right)$$

Experimental Setup:

- Technology: 45 nm STMicroelectronics
- Tools: Synopsys Design Compiler (synthesis), ModelSim (timing analysis), PrimeTime (signal activity)
- Benchmarks: ISCAS85 suite

Simulation Results:

- PG increases initial delay but reduces aging-induced degradation over time.
- For effectiveness, P_{sleep} must be > 0.7 ; otherwise, delay penalties outweigh lifetime gains.
- Parameters (duty cycle), (stress probability), and P_{sleep} (sleep factor) critically influence performance.
- Using $P_{sleep} = 0.8$ yields a significant lifetime extension.
- ST sizing also impacts area overhead and overall efficacy.

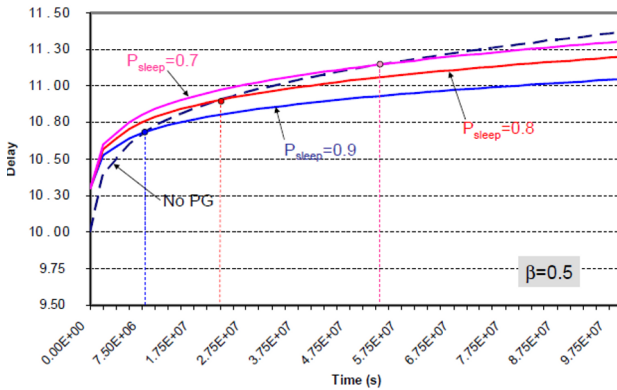


Fig. 1. Delay increase over time with and without power gating

NBTI-Aware Clustered Power Gating

An extension of traditional power gating, clustered power gating offers an aging-aware enhancement by optimizing the

sleep transistor configuration based on criticality analysis of logic paths [7]. Three configurations were evaluated: 1) Full Power Gating: Applies PG to the entire block. 2) Partial Power Gating: Applies PG only to non-critical paths to avoid delay overhead. However, this method shows limited aging benefits, especially for NBTI, due to reduced coverage. Clustered Power Gating: Divides the circuit into critical and non-critical clusters. Critical clusters use larger sleep transistors to minimize delay impact, while non-critical clusters are gated with smaller transistors to maximize area and leakage savings.

The clustered configuration allows fine-grained control of leakage and NBTI-induced degradation while maintaining timing constraints. This technique is particularly effective in multi-core architectures, where each core or functional unit can be individually gated with an appropriately sized sleep transistor.

Advantages: Reduces leakage power and Enables localized lifetime optimization

Drawbacks: Increases area due to additional control and transistor logic

B. Dynamic Voltage and Frequency Scaling (DVFS)

DVFS is a foundational technique in energy-efficient computing systems. It operates by dynamically adjusting both the supply voltage and operating frequency of processing elements to match the required performance level while minimizing power consumption. This dynamic control allows systems to operate in high-performance mode during computation-intensive tasks and switch to lower-power states during idle or low-load periods.

DVFS has become nearly ubiquitous in modern computing—from smartphones to servers—due to its ability to prolong battery life, reduce thermal stress, and enhance overall energy efficiency. In the context of aging mitigation, DVFS also plays a critical role by adjusting operating conditions to compensate for performance degradation caused by aging mechanisms such as NBTI and HCI [8].

Recent research has explored the application of aging-aware DVFS, where the DVFS controller dynamically considers device degradation over time to avoid excessive stress. For example, operating at near-threshold voltages may be energy-efficient, but it can also accelerate aging if not properly managed. Therefore, aging-aware profiles can balance short-term efficiency with long-term reliability.

1) *NBTI-Aware DVFS Technique*: An advanced form of DVFS targets the impact of NBTI through dynamic adjustment of voltage and frequency. The core idea is to mitigate the aging-induced delay increase by adjusting the supply voltage in real-time based on critical path delay and temperature measurements.

The standard NBTI degradation models assume constant voltage and temperature. However, to apply them dynamically, the model is adapted as follows:

$$V_{th}(t', V_2, T_2) = V_{th}(t, V_1, T_1) \quad (1)$$

This formulation allows for recalculating aging effects when voltage and temperature conditions change over time. The system continually measures delay and temperature and uses these to adjust voltage levels while keeping the value below or

equal to the nominal voltage, thus avoiding over-powering and ensuring safe operation.

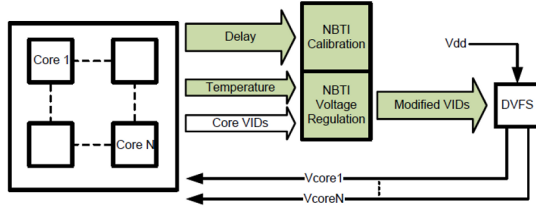


Fig. 2. proposed dvfs architecture

As shown in Fig. 2, the system enhances conventional DVFS with two green-highlighted modules: 1) NBTI Calibration, 2) NBTI-aware Voltage Regulation. These modules use realtime delay and temperature readings to compute the optimal voltage per core, which is then sent to the DVFS unit.

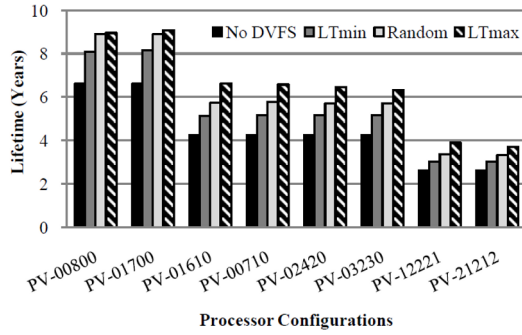


Fig. 3. lifetime with different processors configurations by applying different dvfs techniques

these results show: 1) Using stepped voltage (vs fixed), delay degradation is reduced over time, 2) lifetime is extended significantly using LTmax mapping, where sturdy cores handle the bulk of processing, 3) Dynamic energy savings reach up to 25%, depending on the processor configuration and mapping algorithm, 4) Process variation (PV) scenarios confirm that reducing initial VT mismatch increases both energy efficiency and lifespan as shown in figure 3.

another study is called Core Types & Mapping Algorithms, the study divides cores into: 1) Sturdy cores: Low threshold voltage, long lifespan 2) Frail cores: High threshold voltage, short lifespan

Three mapping strategies were explored: 1) LTMin: Maximize use of frail cores → Short lifespan 2) Random: Uniform, unconstrained task distribution 3) LTMax: Prefer sturdy cores, defer frail usage → Longest lifespan

2) *Alternative DVFS Techniques*: Two additional DVFS-based strategies are evaluated in [9] which operates at elevated voltage levels to complete computations faster. Shuts off supply post-completion to allow recovery from aging. this technique requires process-by-process management, which increases overhead.

[10] applies a lower-than-nominal voltage for recovery periods by using adaptive guardbands based on SPICE simulations and delay measurements. this technique requires ultra-fast voltage regulators and critical path monitors. It Shown to effectively balance aging recovery with energy savings.

Recent work [11] also proposed predictive aging-aware DVFS models using machine learning to anticipate degradation trends — an interesting future direction to integrate with sensor-driven architectures.

C. Input Vector Control (IVC)

IVC is a well-established technique aimed at minimizing leakage current in combinational logic circuits. It leverages the fact that the input values applied to a gate during standby mode directly influence the subthreshold and gate leakage components. By carefully selecting an input vector, it is possible to place transistors into states that exhibit lower leakage and, as a result, also reduce aging.

In the context of NBTI and PBTI degradation mechanisms, IVC has proven useful for limiting transistor stress during standby periods. By selecting a specific vector, the stress on the gate oxide can be mitigated, effectively extending the device's lifetime.

Internal Node Control (INC) [12]: INC was introduced to force internal nodes of logic gates to fixed logic levels (0 or 1) during idle states. This requires additional circuitry, such as multiplexers or transmission gates, which increases chip area and may lead to added delay in the critical path. Three INC variations have been proposed: Gate replacement [13], Gate modification [14] and Transmission gate insertion [15].

These approaches, while effective, are static in nature and require modifications at the design level, making them less practical for post-fabrication aging mitigation.

Using a single static input vector during standby introduces continuous stress on specific transistors, especially in paths involving inverters or inversion logic. This continuous stress accelerates aging in certain parts of the circuit, potentially leading to performance degradation.

a) *Multi-Vector Approach* [16]: To address the limitations of static IVC, a multi-vector strategy was proposed. By cycling through multiple standby vectors, stress can be distributed more evenly across the circuit. However, this requires large memory resources to store the multiple vectors, which can be impractical for memory-constrained designs.

b) *Random Vector Generation* [17]: To reduce memory overhead, [17] introduced a method based on Linear Feedback Shift Registers (LFSRs) to generate pseudo-random input vectors during standby. This method maintains the benefits of distributed stress without requiring extensive storage, and is suitable for runtime implementations with minimal hardware cost.

c) *PVT-Aware Input Vector Selection* [18]: Recent developments have focused on adapting IVC in light of Process, Voltage, and Temperature (PVT) variations, as these parameters significantly influence leakage current and aging rates. [18] proposes an adaptive IVC technique that selects input vectors based on real-time leakage and PVT conditions, improving the aging mitigation effectiveness without sacrificing reliability.

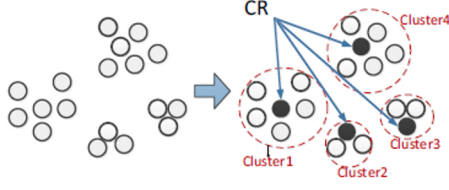


Fig. 4. clustering of critical paths and cluster representatives

D. Adaptive Input Vector Control (A-IVC) [18]

To overcome the limitations of static or random input vector selection, [18] proposes an A-IVC technique that accounts for real-time circuit variations. This method combines IVC with clock gating, and adapts to PVT variations to enhance aging mitigation, especially against NBTI and PBTI effects.

a) *Step 1: Offline Clustering and Characterization:* The technique begins by grouping critical paths with similar aging behaviors (PVT and logic gate overlap) into clusters, where delay variation among paths is highly correlated. For each cluster, Aging-aware timing analysis is conducted, a Pareto-based optimization between NBTI impact and power consumption is used to identify the best input vector, these optimal vectors are stored in Lookup Tables (LUTs) for future use.

b) *Step 2: Online Monitoring and Clock Gating:* At runtime:

- Clock gating is used to disconnect inactive functional units (FUs), minimizing dynamic power.
- A Cluster Representative (CR) path is randomly selected from each cluster.
- Sensors measure real-time PVT values and the delay of the CR.
- The Adaptive Reliability Controller (ARC) uses this data to select the appropriate input vector from the LUT.
- The selected A-IVC is then applied to the entire cluster for aging mitigation.

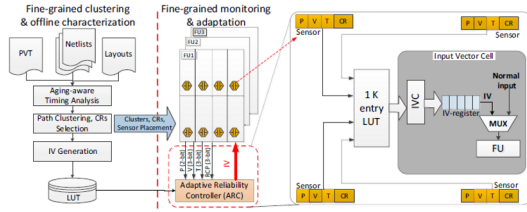


Fig. 5. architecture of monitoring and adaptation system

Fine-grained control allows per-cluster adaptation. Reduces memory overhead by sharing optimal vectors across clustered paths. Enables aging mitigation while maintaining power and area efficiency.

The proposed A-IVC technique was evaluated using industry-grade tools and benchmark circuits: using Synopsys Design Compiler and Cadence SoC Encounter Critical Path Extraction by Synopsys PrimeTime with Clustering Method Elbow method [19] to estimate optimal cluster count

$$PE = \sqrt{\frac{\sum_i (\text{PredictedDelay}_i - \text{ActualDelay}_i)^2}{\text{Delay range}}} \times 100\%$$

Where Delay Range is the difference between maximum and minimum delays at 1.5 and 3 years.

Results and Observations:

K-means clustering method significantly improves delay prediction accuracy compared to both nominal delay estimation and SVD, especially at 1.5 and 3-year marks.

The average lifetime improvement achieved using A-IVC is around 27% compared to Static IVC, as seen in Fig. 8.

Power, area, and delay overheads are minimal (1.3%, 0.9%, and 0.6% respectively), making the technique suitable for practical adoption.

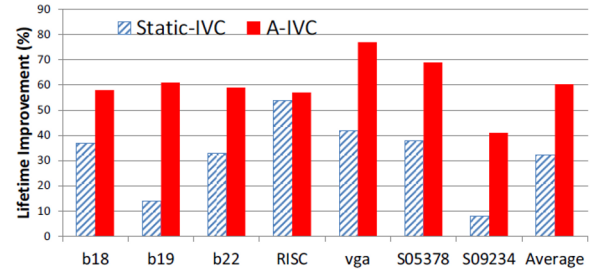


Fig. 6. lifetime improvement comparison between Static-IVC and A-IVC

This technique Achieves 27% average lifetime improvement over Static-IVC

It requires LUTs to store input vectors, Needs additional hardware modification to inject vectors through a multiplexer Clock Gating used in this technique may also reduce HCI, although this effect was not explicitly addressed. The number of cluster representatives (CRs) plays a key role; increasing CRs improves prediction precision and lifetime extension

A novel approach to mitigate NBTI-induced aging at the instruction-level was proposed in [20], leveraging the behavior of NOP (No Operation) instructions. Traditionally, NOPs are used as placeholders in instruction streams; they consume a clock cycle without altering the program state—essentially only incrementing the instruction pointer (EIP).

The technique revolves around the replacement or redefinition of standard NOPs to reduce aging stress: 1) Offline Approach: In the initial static proposal, NOPs generated by compilers are replaced with alternative instructions that are functionally equivalent. 2) Online Adaptation (Proposed Technique): The technique was later extended to support runtime adaptability, making it suitable for dynamic systems exposed to variable PVT conditions.

This approach adds a hardware multiplexer input that allows the processor to dynamically replace a default NOP with a more aging-efficient version, depending on real-time sensor readings. It reduces NBTI stress without changing program logic, doesn't require major software or ISA modification and it is Lightweight and adaptable at the hardware level.

in the other hand, it adds slight complexity to instruction decoding logic, Requires calibration and profiling to identify low-aging instruction alternatives and Beneficial primarily when many idle cycles exist (e.g., in pipelined stalls, branch delay slots)

E. Aging-Aware Task Migration

This method involves redistributing tasks among Processing Elements (PEs) of the same core in a GPGPU system to balance the aging effects such as NBTI.

Countermeasure technique: Task migration between PEs + Power gating. The method proposed in [21] includes a two-level system: 1) Software layer: Dynamic Binary Optimizer (DBO) and 2) Hardware layer: Power gating and NBTI delay sensors

The DBO runs on the host CPU and simulates the expected threshold voltage shift (V_{sub}, th_{sub}) for each PE, based on the predicted usage. It then: 1) Measures actual post-silicon degradation, 2) Calibrates prediction using real data, 3) Re-orders PEs from most to least aged, 4) Migrates instructions to balance aging and 5) Regenerates a “healthy” binary

IV. EVALUATION PLATFORMS FOR AGING COUNTERMEASURES

To determine the most suitable and effective aging mitigation strategy for a given system, various countermeasure techniques must be evaluated under realistic aging conditions. Several research efforts have proposed simulation platforms that accurately model aging mechanisms such as NBTI and HCI, along with environmental parameters like temperature and power consumption. This section presents an overview of the ExtraTime platform [22], which evaluates different countermeasures using microarchitectural-level simulations.

A. ExtraTime: Microarchitectural-Level Aging Modeling Platform

The ExtraTime platform [22] allows simulation-based evaluation of three countermeasure strategies: Power Gating, Clock Gating, and Instruction Scheduling. The platform supports aging models based on both NBTI and HCI mechanisms using a 32 nm technology process. It targets mono-core systems and monitors online parameters such as temperature and critical path delay.

1) *Evaluation of Execution Unit Policies:* Five different scheduling policies were evaluated to analyze their impact on aging and performance (see Fig. 8):

Policy	Description	Cycle: Instructions:	Cycle 1 A	Cycle 2 B,C	Cycle 3 D,E	Cycle 4	Cycle 5
Stat1	Always use the same EU		EU1 (A)	EU1 (B)	EU1 (C)	EU1 (D)	EU1 (E)
Stat2	Always use the same EU		EU1 (A)	EU1 (B)	EU1 (C)	EU1 (D)	EU1 (E)
Stat2	If this one is not available, take another one		EU1 (A)	EU2 (C)	EU2 (E)		
CQ1	Change EU with every new instruction		EU1 (A)	EU2 (B)	EU2 (D)	EU1 (E)	
CQ1	If unit is not available try next one		EU1 (A)	EU1 (C)	EU1 (E)		
CQ2	Change EU only, if it is not available		EU1 (A)	EU1 (B)	EU2 (D)	EU1 (E)	
CQ2	Always use the same EU		EU1 (A)	EU2 (C)	EU1 (E)		
CQ3-X	Change EU every X cycles (here: X = 3)		EU1 (A)	EU1 (B)	EU1 (C)	EU2 (D)	EU2 (E)

Fig. 7. scheduling policies and their impact on aging and performance

2) *Integration of Temperature and Power Modeling:* Given the direct relationship between power consumption and thermal behavior, the ExtraTime platform tightly couples power and thermal models. Accurate temperature estimations are integrated into the aging models, enhancing prediction accuracy for both NBTI and HCI-induced degradations.

3) *Power Gating Configuration Optimization:* The simulator searches for optimal idle duration (T_{idle}) and gating period (T_{gating}) configurations to minimize aging while maintaining performance. These optimizations are based on the estimated thermal and stress parameters under realistic workloads.

B. On the Efficacy of NBTI Mitigation Techniques [23]

A numerical model of threshold voltage degradation in PMOS transistors due to Negative Bias Temperature Instability (NBTI) was proposed in [23]. This model assumes uniform aging across all PMOS transistors within a microarchitectural block. Unlike traditional analytical models, this approach accommodates variations in supply voltage over time, offering a more realistic prediction of degradation behavior.

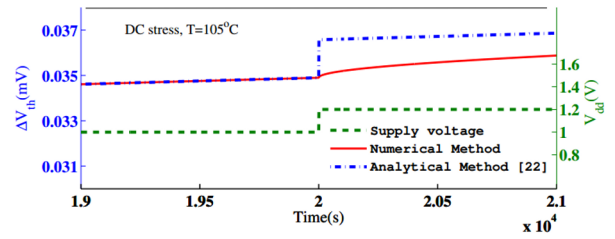


Fig. 8. comparison between the numerical and analytical methods under a DC stress condition

Figure 7 illustrates the comparison between the numerical and analytical methods under a DC stress condition at 105°C. The numerical model (red curve) shows a gradual adaptation to the change in supply voltage, whereas the analytical method (blue dashed curve) responds instantaneously, which can lead to inaccuracies in scenarios involving dynamic voltage scaling.

This improvement in modeling accuracy provides a foundation for designing more adaptive and resilient aging-aware systems.

1) *Evaluation of NBTI Mitigation Techniques:* The authors in [23] evaluate the effectiveness of multiple aging mitigation strategies using detailed simulations and modeling tools. The following techniques were analyzed: 1) DVFS, during simulations, SPICE was used to measure the delay of critical paths selected via Static Timing Analysis (STA). If the delay of any critical path exceeds the defined guardband relative to the operating frequency, the supply voltage is increased by 5 mV. The results show that DVFS can effectively reduce the overall energy consumption of the chip by adapting the voltage dynamically based on aging-induced delay degradation. 2) PG, the evaluation demonstrates that the effectiveness of Power Gating is highly dependent on its configuration. Specifically, the duration for which power gating is applied plays a key role. Longer power gating periods lead to a smaller guardband

in frequency margins, enabling a higher operating frequency and therefore improved performance. This highlights a tradeoff between energy savings and performance recovery through intelligent PG tuning. 3) Activity Management, in contrast to the two previous techniques, variations in the activity factor were found to have little to no impact on NBTI-induced aging. This observation stems from the complementary nature of CMOS technology and the inherent signal inversion in combinational circuits. Consequently, toggling activity does not significantly influence the average stress experienced by PMOS transistors under NBTI.

V. COMPARISON OF EXISTING TECHNIQUES

Table 2 provides a qualitative comparison of existing NBTI mitigation techniques across different levels of design abstraction: gate-level, RTL, and system-level. Three primary abstraction levels are identified:

- Gate Level: Offers high accuracy and technology-dependent timing modeling, but with a very slow simulation speed. It is primarily used for chip fabrication.
- Register Transfer Level (RTL): Modeled using hardware description languages such as VHDL or Verilog, RTL-level models are suitable for automatic transformation into physical designs. However, simulations are also slow due to their detail.
- System Level: Uses Transaction-Level Modeling (TLM) to represent the system architecture with concurrent components and communication channels. It is widely adopted for SoC prototyping, offering faster simulation and reduced modeling effort due to its high abstraction.

Each technique addresses NBTI and/or HCI degradation using either software-level or hardware-level countermeasures, as shown in Table 2.

A. Summary of Countermeasure Techniques

- Power Gating: Highly efficient but requires hardware modification. Operates at RTL and system level.
- DVFS: Also highly efficient with system-level implementation, but requires an integrated voltage regulator.
- IVC: Offers moderate efficiency without performance impact but increases memory requirements.
- Instruction Scheduling: Provides high performance but demands at least two functional units (FUs) of the same type per core.
- Task Migration: Balances threshold voltage mismatch across PEs/cores but requires system shutdown for re-implementation and is difficult to apply.

B. Conclusion

This paper surveys various online and offline countermeasures targeting transistor aging effects such as NBTI and HCI in mono- and multi-core systems. The proposed mitigation strategies span across:

- Software-based approaches (e.g., instruction and task migration between cores/PEs),
- Hardware-based approaches (e.g., NOP-based instruction scheduling, IVC),

- Hybrid techniques (e.g., DVFS, Power Gating).

While techniques like task migration and instruction scheduling are powerful, they are not general-purpose and may only suit well-defined embedded systems or specific applications. In contrast, methods like DVFS and PG are more broadly applicable across platforms like smartphones or general-purpose computers.

Limitations remain in the current state-of-the-art, particularly regarding online aging monitoring. Most techniques rely on offline analysis and are not yet adaptive to dynamic system states. Future work should aim to integrate real-time aging sensors and adaptive models that respond to actual operating conditions.

VI. PROPOSED ONLINE MITIGATION APPROACH

A. Objective

The objective of this approach is to implement online aging mitigation techniques on a heterogeneous multiprocessor platform, with two main goals: 1) Minimize architectural changes, ensuring low hardware overhead, 2) Minimize performance impact, maintaining application throughput.

This is achieved by introducing a Smart Monitoring Controller (SMC) that enables real-time observation and control of mitigation techniques such as DVFS, task migration, and power gating.

B. Target Architecture: STELLAR Platform

The selected case study is the STELLAR platform, a RISC-V-based heterogeneous multicore system following a big.LITTLE architecture as shows in figure 9.

Key Features: 1) 4 cores: 2 BOOM (Out-of-Order, superscalar) + 2 Rocket (In-Order). 2) Separate power and clock domains for Rocket and BOOM clusters. 3) Uses RISC-V ISA, supporting simulation at RTL (ModelSim/VCS) and TLM (SESAM).

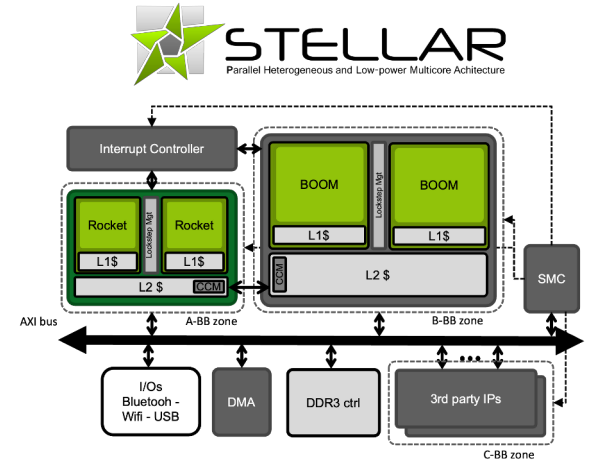


Fig. 9. Overview of the STELLAR platform architecture with BOOM and Rocket cores in separate power/clock domains

TABLE II
COMPARISON OF AGING COUNTERMEASURE TECHNIQUES

Technique	Mechanism	Advantage	Disadvantage	Parameter	Level	Tech (nm)
Power Gating [6], [7], [13], [15]	NBTI	High efficiency	Circuit modification	Voltage	RTL, system	45, 90
DVFS [8], [10], [11]	NBTI	High efficiency	Requires voltage regulator	Voltage, Temperature, Delay	System	45, 22
IVC [12], [14], [16], [17], [24]	N/PBTI, HCI	Moderate efficiency; No performance impact (NOP)	Memory overhead	PVT, CP delay	Gate level	45, 65, 90
Instruction Scheduling [20]	NBTI, HCI	High performance	Requires at least two same-type EUs per core	VT, CP delay	System	90
Task Migration [18], [25]	NBTI	Reduces threshold mismatch between PEs or cores	System halt during code re-implementation; Complex to apply	Voltage, CP delay	System	65, 22

C. Aging Mitigation Management via SMC

We propose a dedicated microcontroller, the Smart Monitoring Controller (SMC), responsible for collecting aging-related metrics and orchestrating mitigation actions as illustrated in figure 10.

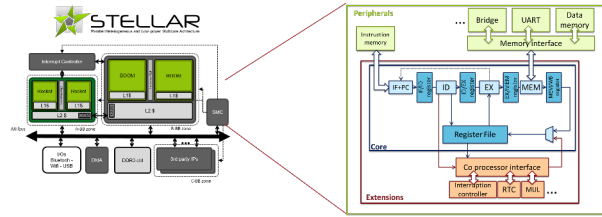


Fig. 10. Internal architecture of the Smart Monitoring Controller (SMC) and its integration with STELLAR

Each core is equipped with three sensors: Temperature (T), Volatage (V) and Degradation (D). These sensors provide real-time data to the SMC for analysis.

The OS (running on Rocket cores) informs the SMC of core activity (active/inactive) and task type. The SMC, based on sensor feedback, may request task migration if aging becomes unbalanced across cores.

The SMC issues commands to external voltage/frequency regulators and power switches. This enables real-time DVFS and selective power gating of idle cores.

D. Global System Flow

The complete interaction between all components is shown the figure 11:

E. Implementation Notes

The SMC logic is hosted on a dedicated core: 1) using 32-bit RISC Harvard Architecture, 2) In-order, 5-stage pipeline Named AntX

All aging mitigation logic runs in software on the AntX core. Modular software design: each mitigation technique is implemented as a separate module (DVFS, Scheduler, PG).

F. Scheduling Policy for Aging Mitigation

To efficiently distribute workload and extend system lifetime, the SMC includes a custom aging-aware scheduler. This scheduler prioritizes: 1) Assigning *light-weight or less-critical* tasks

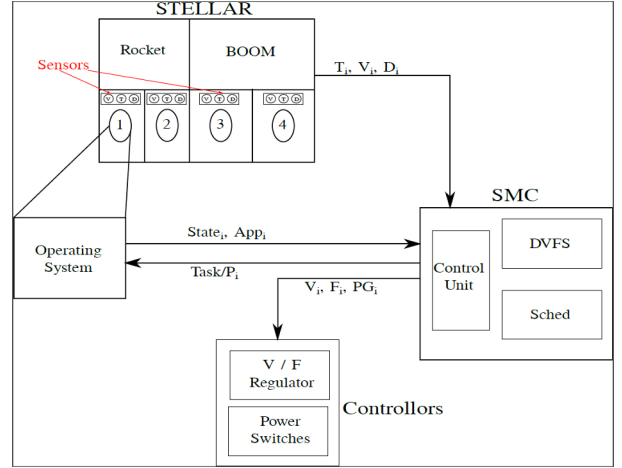


Fig. 11. Global data flow: Sensors send data to the SMC; the SMC interacts with the OS and controls external regulators

to aged cores, 2) Migrating *critical tasks* to less-aged or faster cores (e.g., BOOM instead of Rocket), 3) Triggering power gating for idle cores exceeding an inactivity threshold, and 4) Dynamically adjusting core voltage/frequency using DVFS based on real-time slack analysis.

These decisions rely on degradation metrics D_i from each core's sensor bank.

G. Planned Simulations and Evaluation Strategy

Although no experimental results are available yet, the approach will be validated via multi-level simulation, combining both RTL and TLM models: 1) RTL Simulation for accuracy using ModelSim and VCS it simulates and evaluates the sensor behavior, Power gating response delay and it has Micro-architectural impact on critical paths. 2) TLM Simulation for performance using SESAM simulator. it is faster simulation cycles and Allows simulation of longer execution traces with Acceptable accuracy (~20-30% error tolerance)

The baseline will be the STELLAR system without mitigation. Each technique (DVFS, task migration, PG) will be evaluated individually and combined.

VII. CONCLUSION AND FUTURE WORK

In this paper, we presented a novel aging-aware runtime mitigation framework targeting heterogeneous multicore architectures, specifically applied to the STELLAR platform. By integrating a SMC, our system performs online aging monitoring using temperature, voltage, and degradation sensors, and applies mitigation actions such as task migration, DVFS, and power gating.

This approach offers two major benefits: 1) Modularity – Each mitigation policy is implemented as an independent module within the SMC, facilitating integration and extensibility. 2) Low Intrusiveness – The techniques are deployed with minimal architectural changes, ensuring that performance overhead remains low while improving component lifetime.

We also introduced a new scheduling policy based on real-time degradation ranking that dynamically adapts task allocation and resource control, with the goal of balancing wear across cores while preserving application performance.

Although the implementation and simulation results are still in progress, we have laid the foundation for a comprehensive evaluation strategy using ModelSim/VCS at RTL level and SESAM at TLM level.

By enabling adaptive aging mitigation in real time, this work aims to make aging-aware design practical and scalable for modern multiprocessor systems-on-chip (MPSoCs).

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